



Beyond Input Attribution: A Hands-On Tutorial to Concept-Based Explainable AI and Mechanistic Interpretability

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Meet the instructors and contributors



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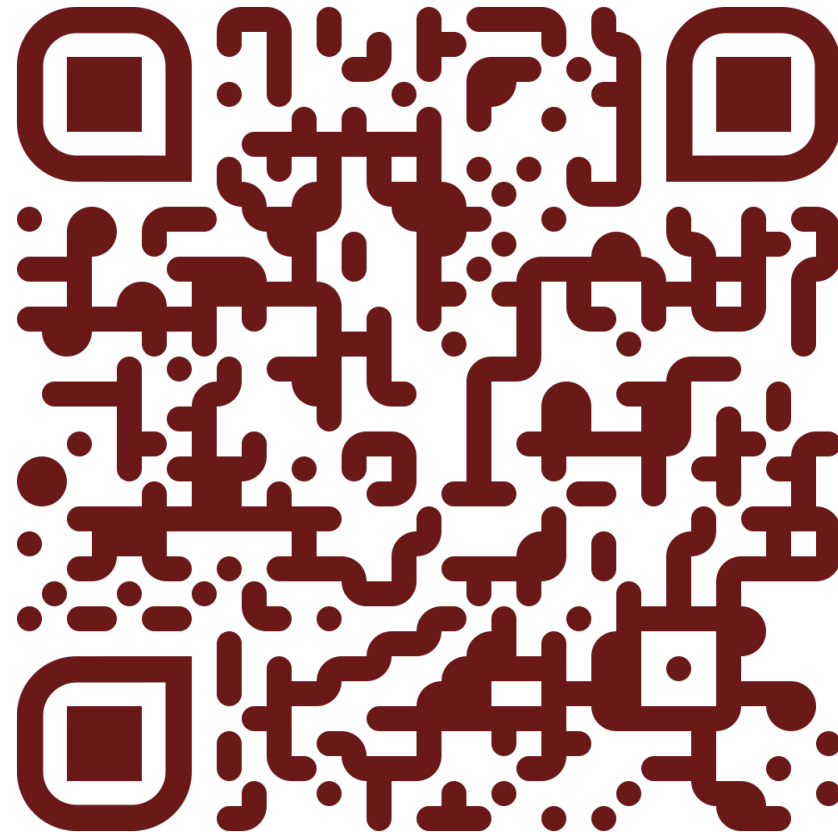
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Check our website for all the material!

<https://cxai-mechint-htutorial-kdd2025.github.io/>



Hands-on tutorial structure

- Introduction to Explainable AI (XAI)
- Concept-Based XAI (C-XAI)
- Mechanistic Interpretability
- Wrap-up and open discussion



Hands-on tutorial structure

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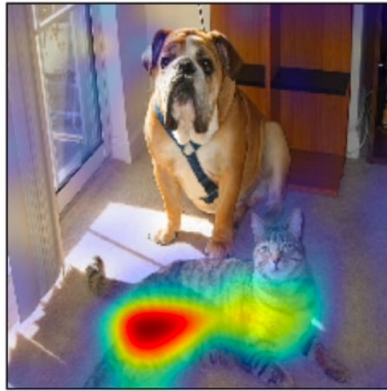
Why Explainability Matters in AI

- Deep learning systems are everywhere
- These systems influence critical decisions
- Demand is growing for AI that is **trustworthy** and **understandable**



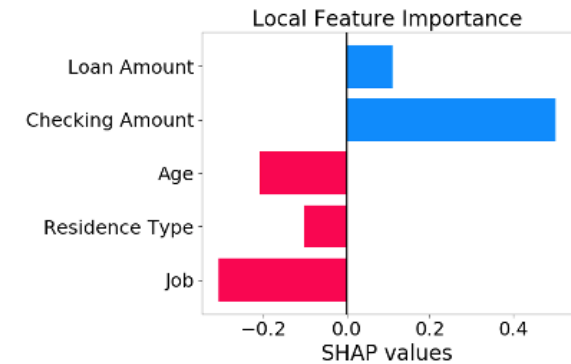
Multiple XAI solutions proposed

Input attributions



(c) Grad-CAM 'Cat'

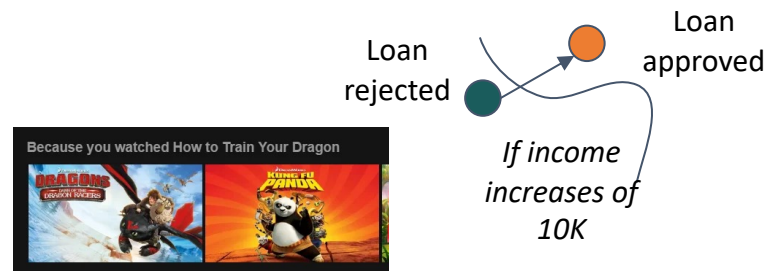
I am really happy



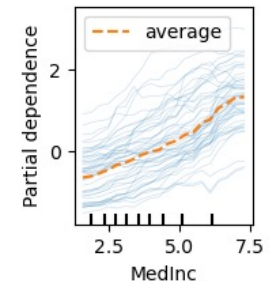
Local rules

If Country is US, married, work hours > 45 $\rightarrow$ Income > 50K

Explanations by example



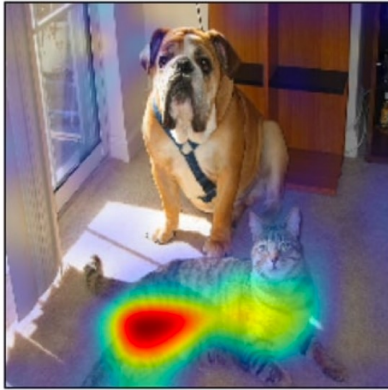
Visualizations



Multiple XAI solutions proposed

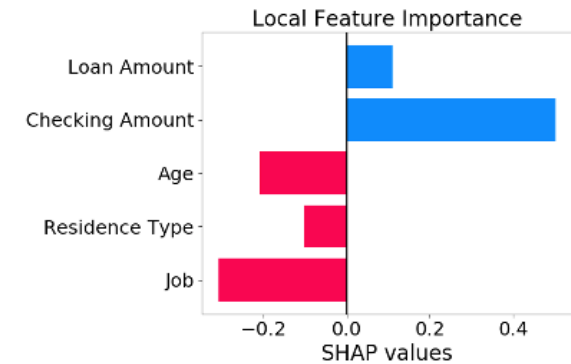
Especially of this category

Input attributions



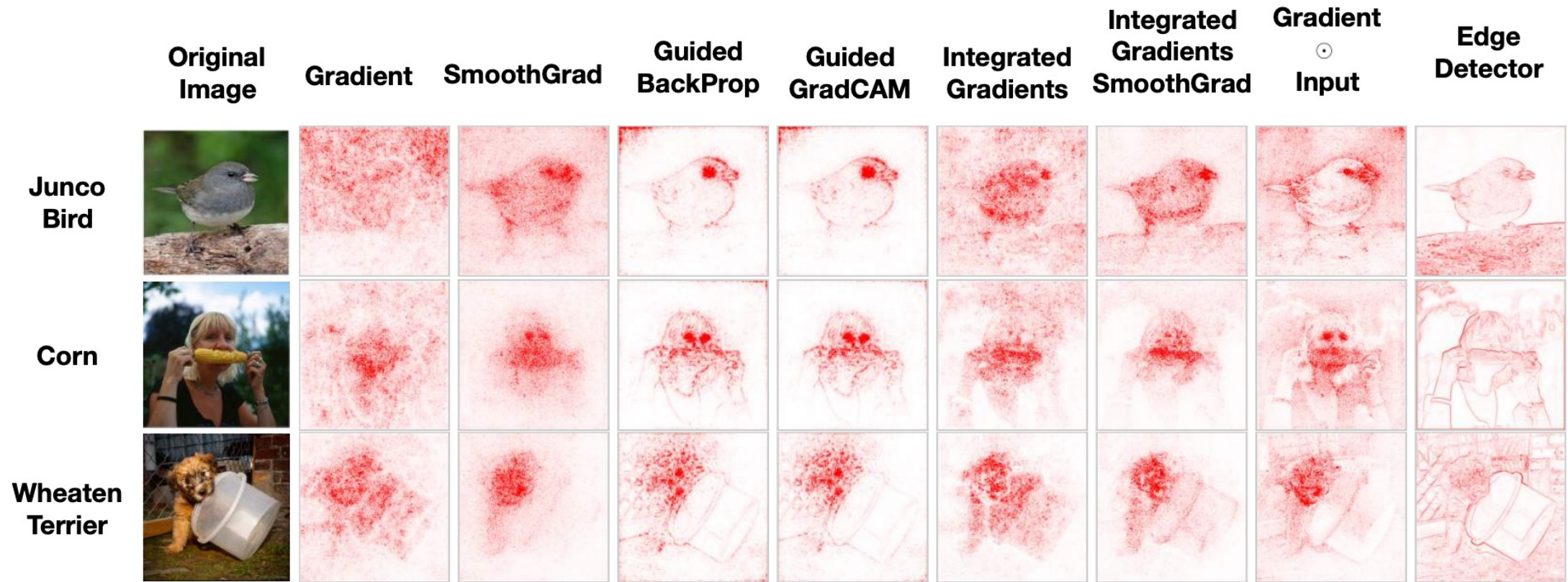
(c) Grad-CAM 'Cat'

I am really happy



Identify which input features (e.g., pixels, words, attributes) contribute most to the model's output

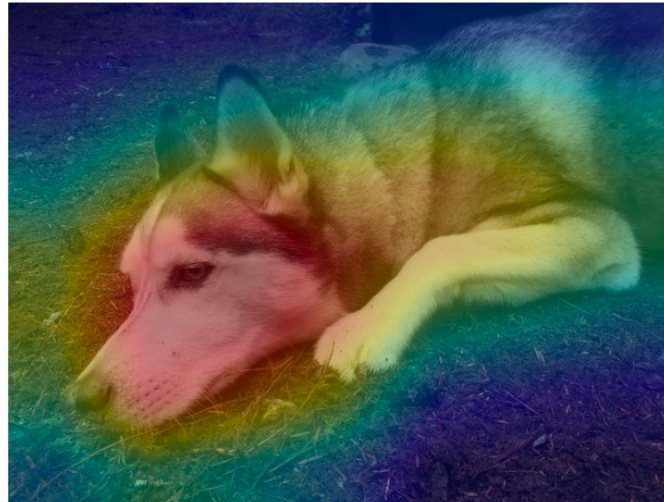
Standard Explainable AI does not always work well - Edge detectors?



Standard Explainable AI does not always work well - Which class are we explaining?



Evidence for Siberian Husky



Evidence for Transverse Flute



The **where** vs the **what**

Showing **where** a network is looking
does not tell us what the network is seeing
in a given input

Beyond Input Attribution: Two Interpretability Paradigms

Concept-Based XAI (C-XAI)

Explain model behavior via
**high-level, human-
interpretable concepts**

Mechanistic Interpretability

Analyze the **internal structure**
of neural networks

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What is a concept?

A concept can be any abstraction, such as a colour,
an object, or even an idea

Types of concepts

- **Symbolic Concepts**
Human-defined attributes
- **Unsupervised Concept Basis**
Cluster of similar samples
- **Prototypes**
(Part-of) a training sample
- **Textual Concepts**
Textual representation

“BEAK”

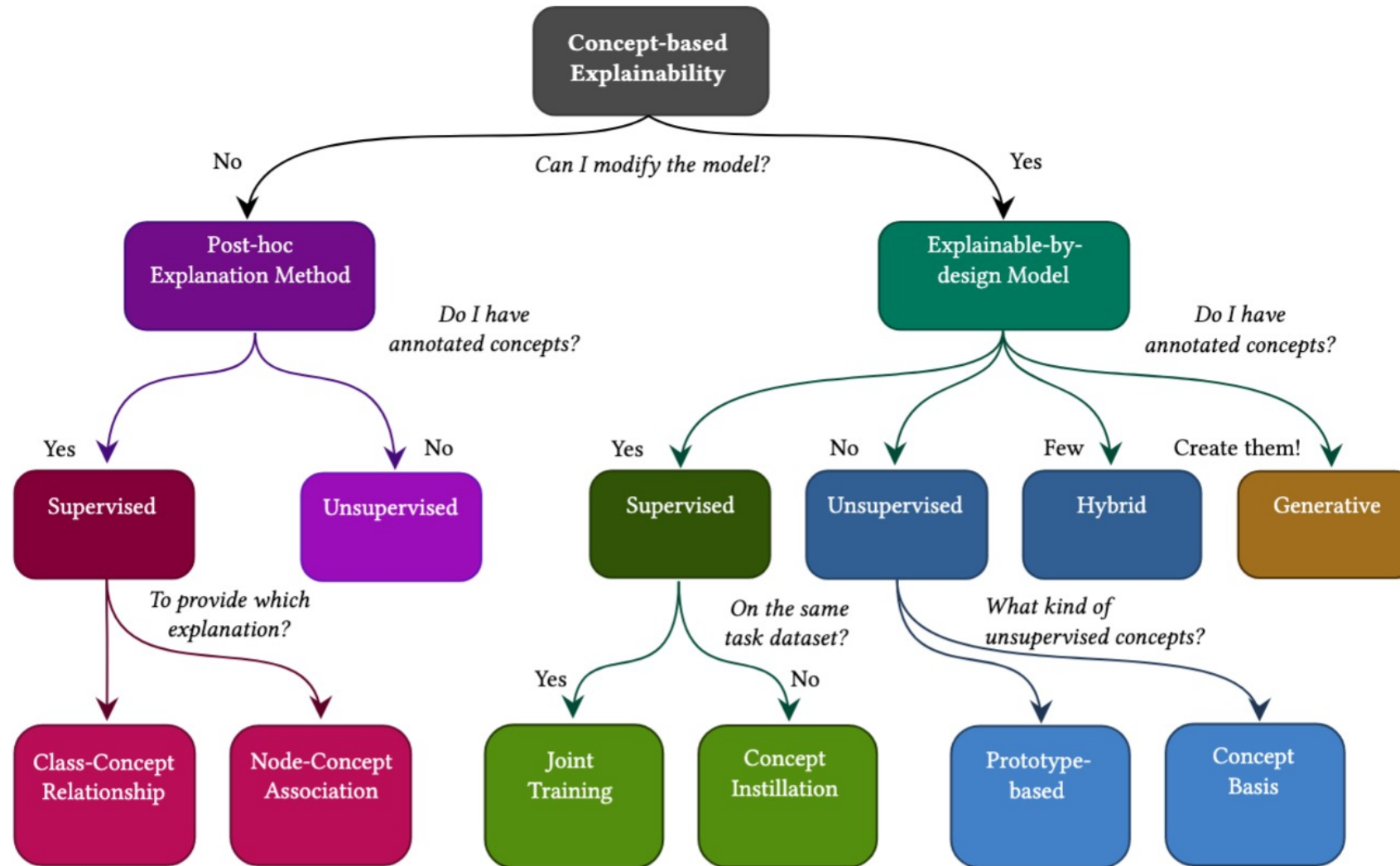
“GREEN”

“WINGS”

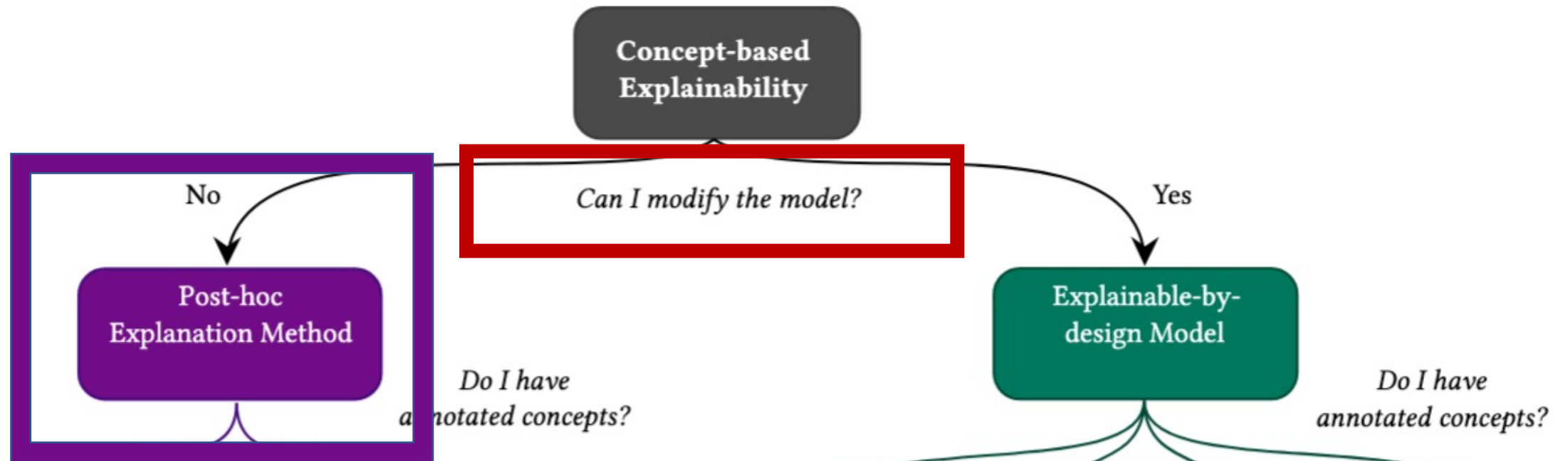


“A bird with bright feathers”

C-XAI Taxonomy



C-XAI Taxonomy



Post-hoc Concept-based Explanation methods

Explain via concepts an **already trained model** that we cannot (or do not want to) modify

Often, the approaches are based on:

- Project samples representing the concepts in the model latent space
- Analyze their relationship to the prediction (or the hidden node activations)

Post-hoc Concept-based Explanation methods

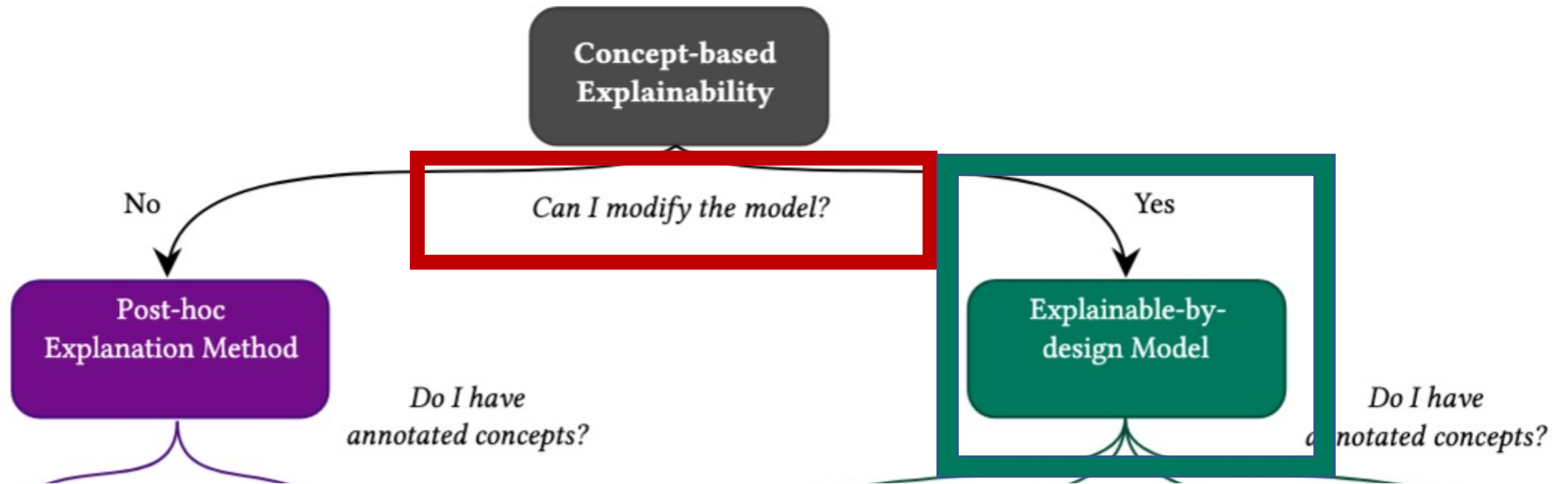
Pros.

- They **do not compromise the learning capacity** of a model
- They can provide more interpretable explanations than standard post-hoc methods

Cons.

- **Cannot ensure the network really knows the concepts**
 - It has not been trained for that
 - Not interpretable by design

C-XAI Taxonomy



Explainable-by-design Concept-based Models

Directly **train concept-aware models**

- Neural models with an explicit concept representation, typically as an intermediate layer
- Predicted concepts influence the task predictions

Explainable-by-design Concept-based Models

Pros.

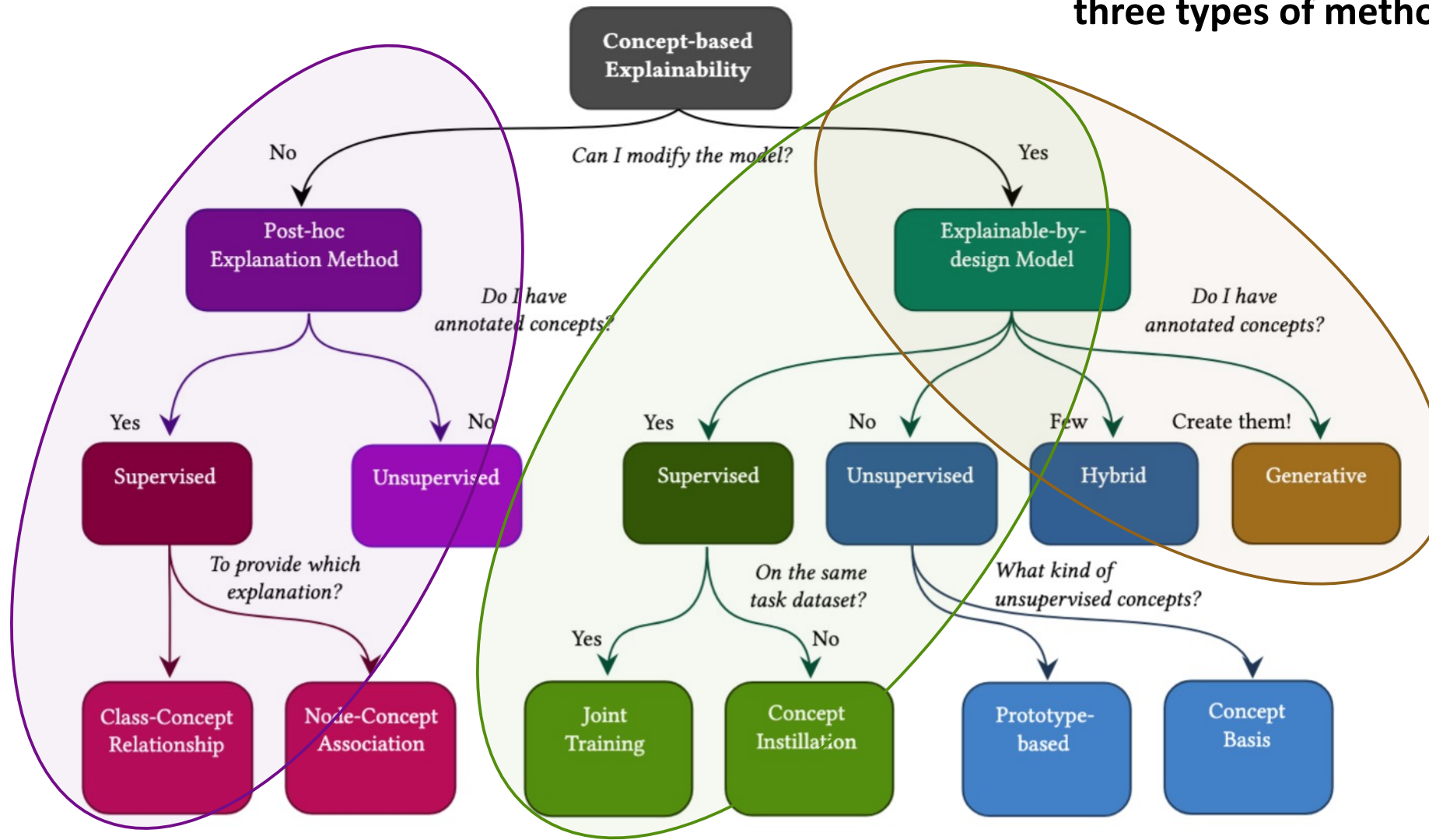
- They can be regarded as **inherently transparent models** as they provide node-concept association by-design

Cons.

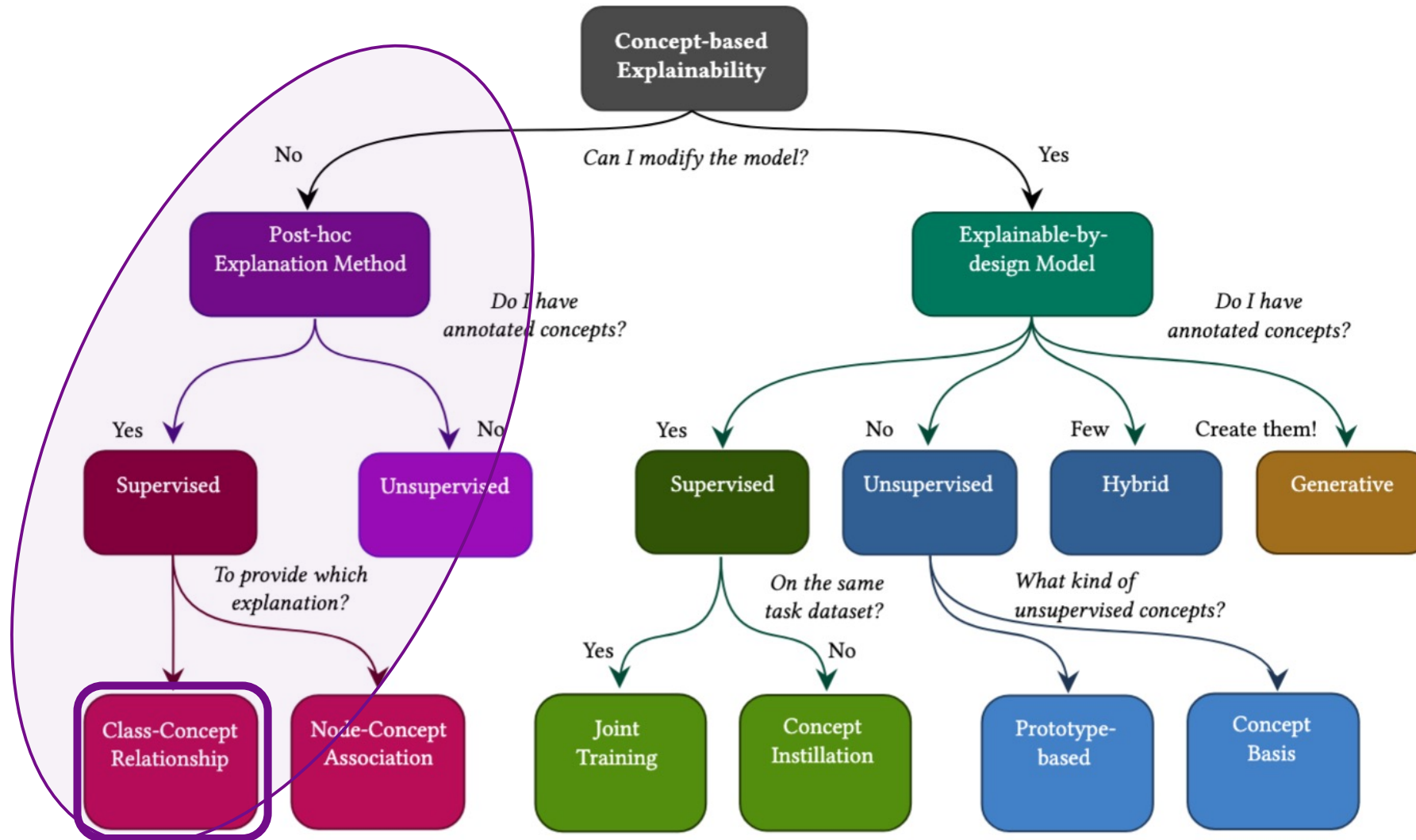
- They need ad-hoc training
- Predicting concepts **might reduce network task performance**

C-XAI Taxonomy

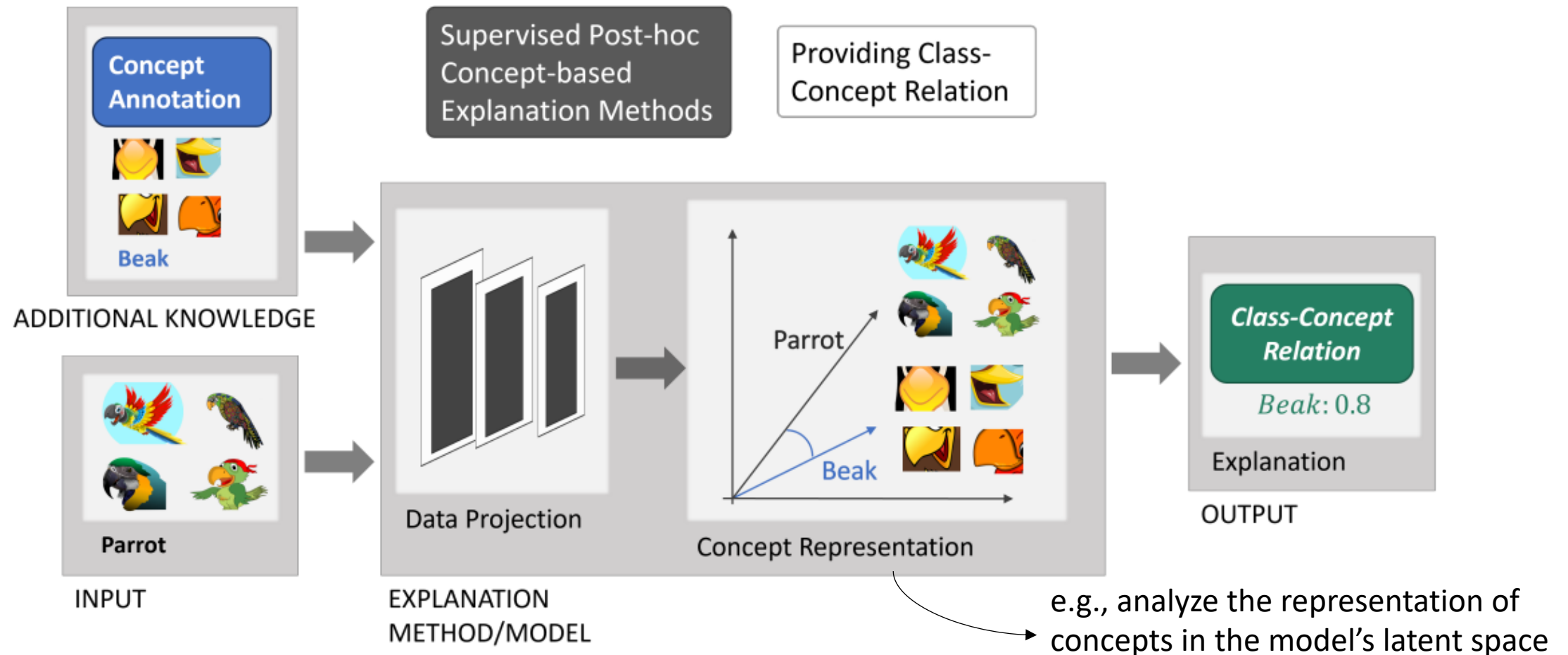
**In this tutorial, we will cover
three types of methods!**



C-XAI Taxonomy



Post-hoc Supervised Methods providing class-concept relations



Testing with Concept Activation Vector (T-CAV)

Concept data: (i) user-defined set of examples for a **concept** (e.g., 'striped'), and **random** examples.

(a)

'Striped' concept

Random



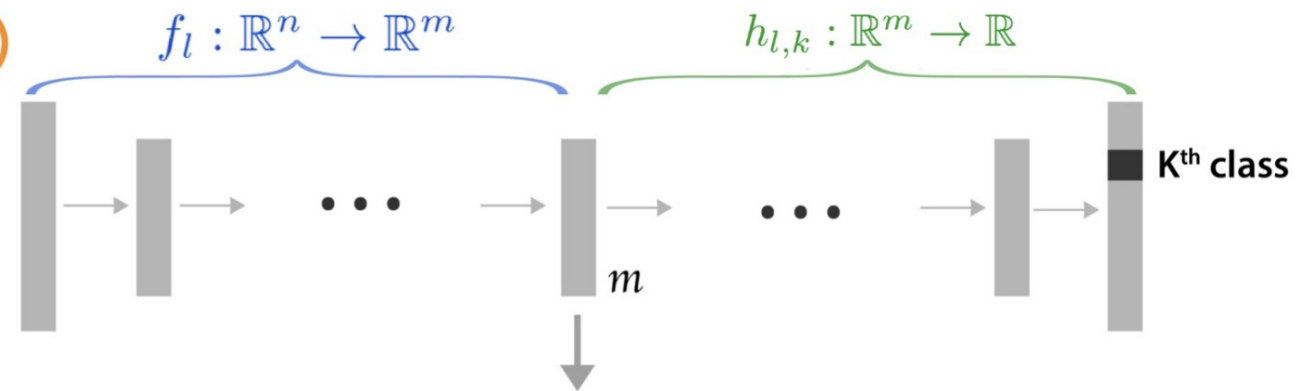
(b)



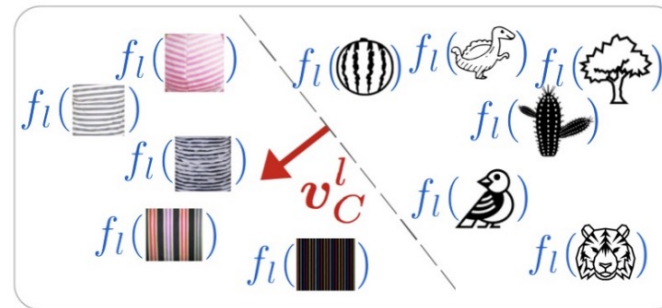
Labeled **training data** examples for the studied class (zebras).

Trained network we want to explain.

(c)



(d)



Train a **linear classifier** to distinguish between the activations produced by a **concept's** examples and **others** in layer l .

The **CAV** v_C^l is the **vector orthogonal** to the classification boundary.

(e)

$$S_{C,k,l}(\text{zebra}) = \nabla h_{l,k}(f_l(\text{zebra})) \cdot v_C^l$$

TCAV uses the **directional derivative** $S_{C,k,l}(x)$ to quantify **conceptual sensitivity** for a given class of interest (zebras).

TCAV – Definition of Conceptual Sensitivity

Let v_c^l be the CAV vector for a concept C in layer l and $f_l(x)$ the activations for input x at layer l

The **conceptual sensitivity $S_{C,k,l}(x)$** is the **directional derivative**

$$S_{C,k,l}(x) = \lim_{\epsilon \rightarrow 0} \frac{h_{l,k}(f_l(x) + \epsilon v_c^l) - h_{l,k}(f_l(x))}{\epsilon} = \nabla h_{l,k}(f_l(x)) \cdot v_c^l$$

$S_{C,k,l}(x) > 0 \rightarrow$ positive influence of concept C

$$\begin{aligned} S_{C,k,l}(\text{zebra}) \\ = \nabla h_{l,k}(f_l(\text{zebra})) \cdot v_C^l \end{aligned}$$

TCAV score

Let k be a class label of interest (e.g., zebra) and X_k the inputs with that label.

The **TCAV score is the fraction of k -class inputs** whose l -layer activation vector was **positively influenced by concept C** , i.e., having a positive directional derivative w.r.t. the CAV

$$TCAV_{Q_{C,k,l}} = \frac{|\{x \in X_k : S_{C,k,l}(x) > 0\}|}{|X_k|}$$

$$\begin{array}{l} S_{C,k,l}(\text{zebra}) \\ S_{C,k,l}(\text{zebra}) \\ S_{C,k,l}(\text{zebra}) \\ S_{C,k,l}(\text{zebra}) \end{array} > 0?$$

TCAV score example

Is the **stripe concept** important to classify 🦓 class for the ML model?

Concept set -
stripes images



Labeled training-data
examples for the
zebra class



$$S_{C,k,l}(\text{zebra})$$

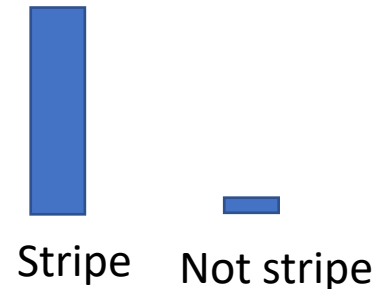
$$S_{C,k,l}(\text{zebra head})$$

$$S_{C,k,l}(\text{zebra body})$$

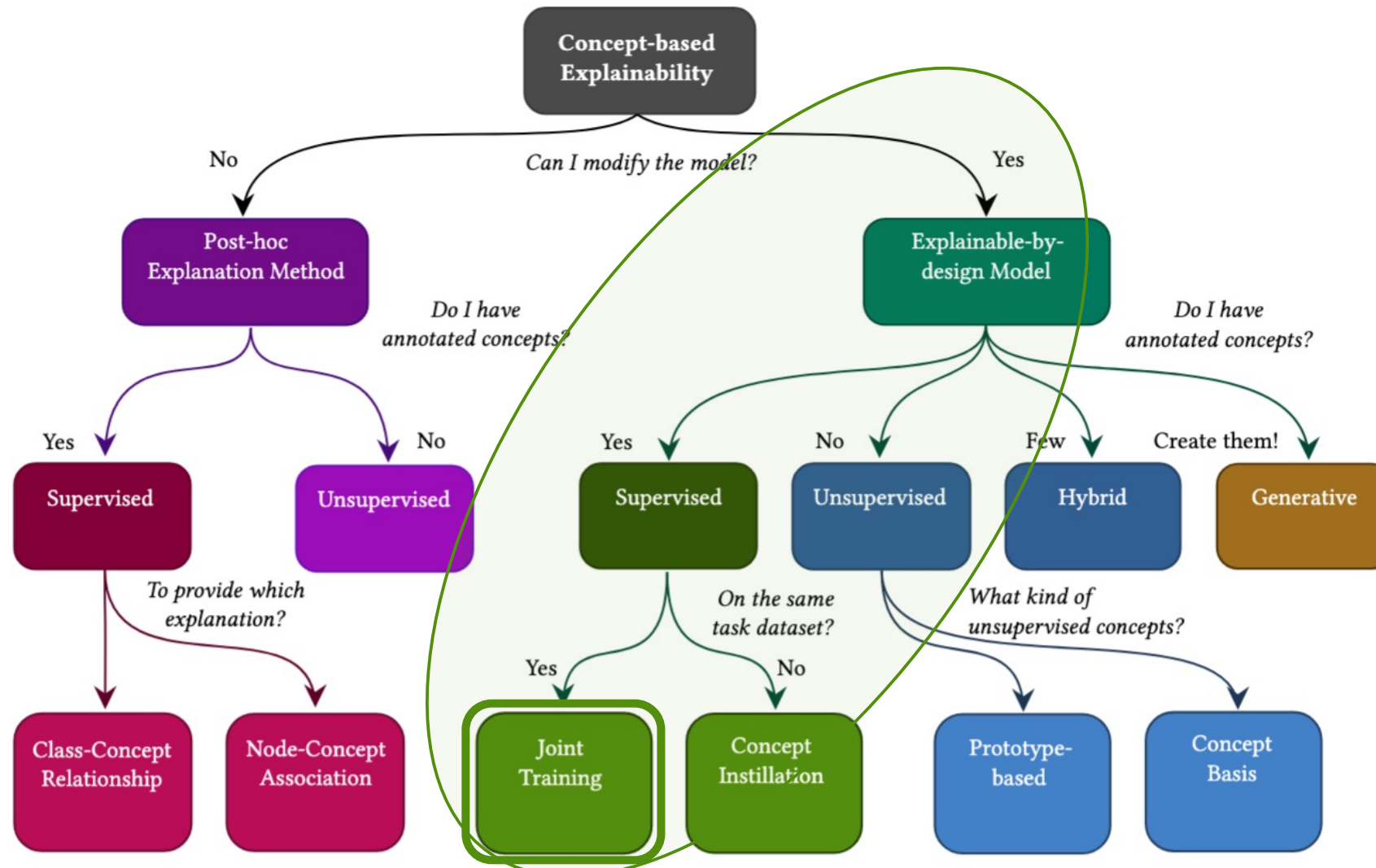
$$S_{C,k,l}(\text{zebra head/body})$$

>0?

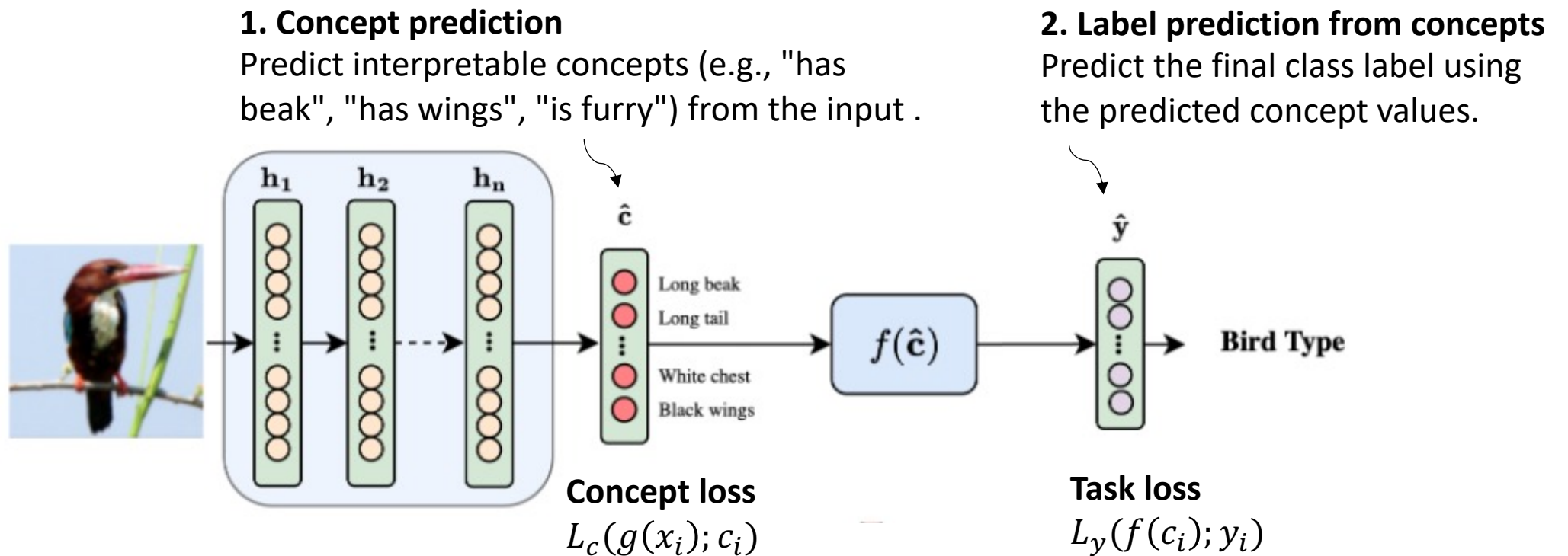
TCAV tells that Stripe has a positive
importance for the classification of zebras



C-XAI Taxonomy



Concept Bottlenecks

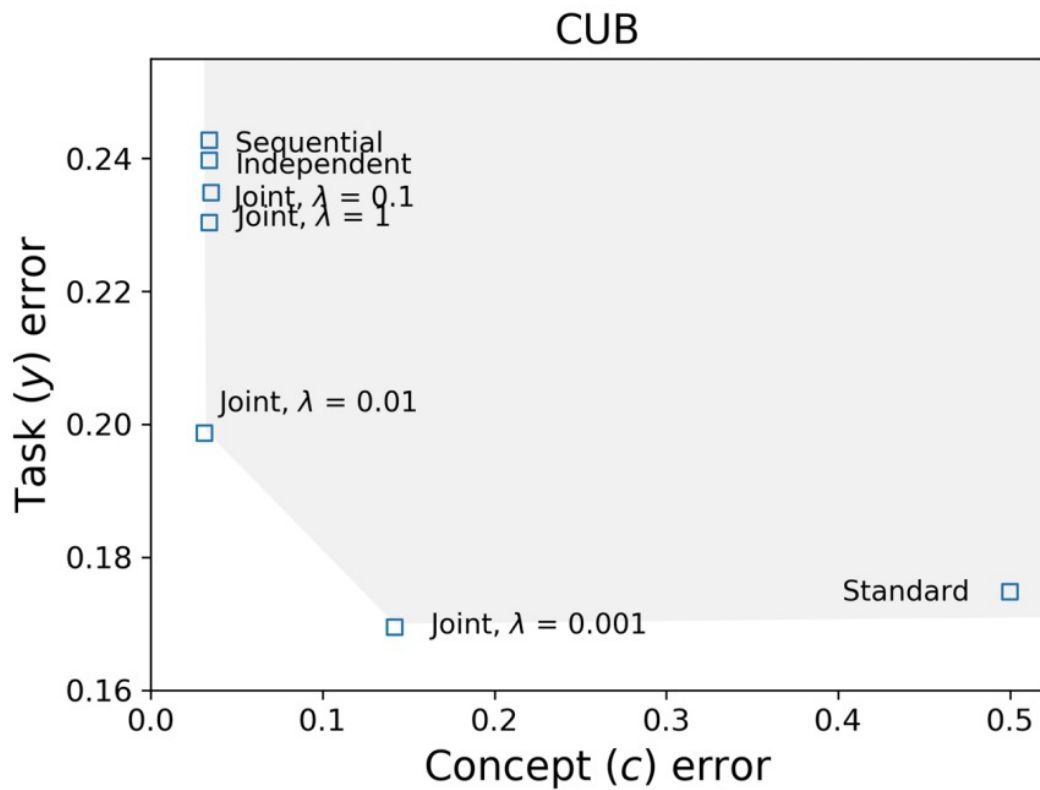


Concept Bottleneck because all information used for prediction must pass through the concept layer — the bottleneck.

Different training strategy

- **Independent** $\hat{f} = \arg \min_f \sum_i L_y(f(c_i), y_i)$ $\rightarrow f$ is trained using the truth concepts
 $\hat{g} = \arg \min_g \sum_i L_c(g(x_i), c_i)$
- **Sequential** $\hat{f} = \arg \min_f \sum_i L_y(f(g(x_i)), y_i)$ $\rightarrow g$ is trained (as above), then freezed
- **Joint** $\hat{f}, \hat{g} = \arg \min_f \sum_i L_y(f(c_i), y_i) + \lambda \arg \min_g \sum_i L_c(g(x_i), c_i)$ $\rightarrow f, g$ trained together for some $\lambda > 0$
- **Standard** $\hat{f} = \arg \min_f \sum_i L_y(f(c_i), y_i)$ $\rightarrow$ It ignores the concepts loss

Interpretability/performance trade-offs

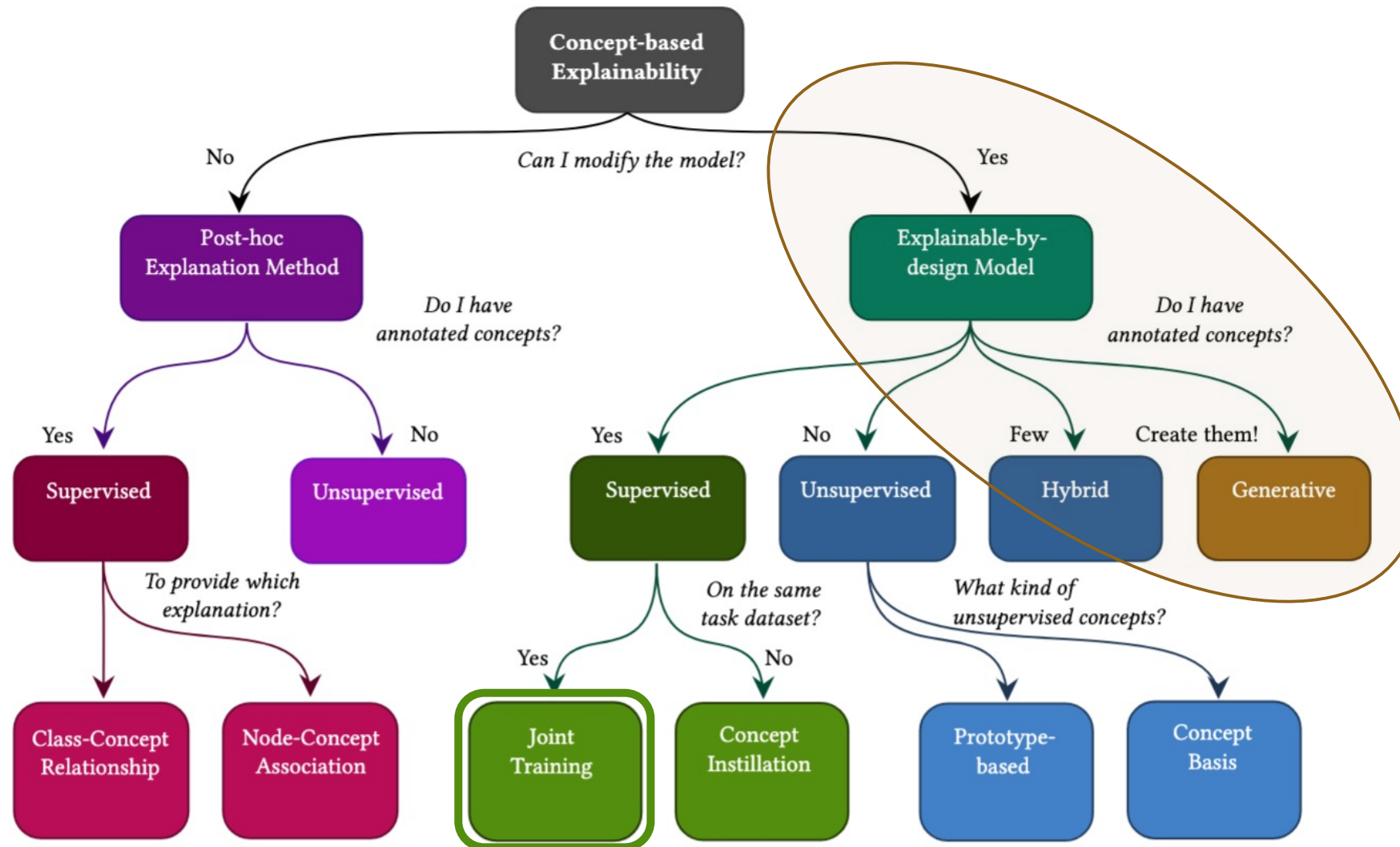


- **Independent** and **Sequential** more 'trustworthy' as they ensure no concept leakage
- **Joint strategy** provides better task accuracy
 - Different trade-offs according to the λ value
- Standard model still has higher accuracy on average
 - But no interpretability

Limitations of Concept Bottlenecks

- Concepts are represented with supervised scalars
 - Solution (as in CEM): concept embeddings, i.e., represent and encode concepts as continuous vectors in semantic space
- **Require labeled data for concepts**
 - Time consuming, costly
 - Fixed set, not flexible, automated
 - Among solutions: prototype-based models, generative C-XAI as label-free which generate concepts

C-XAI Taxonomy



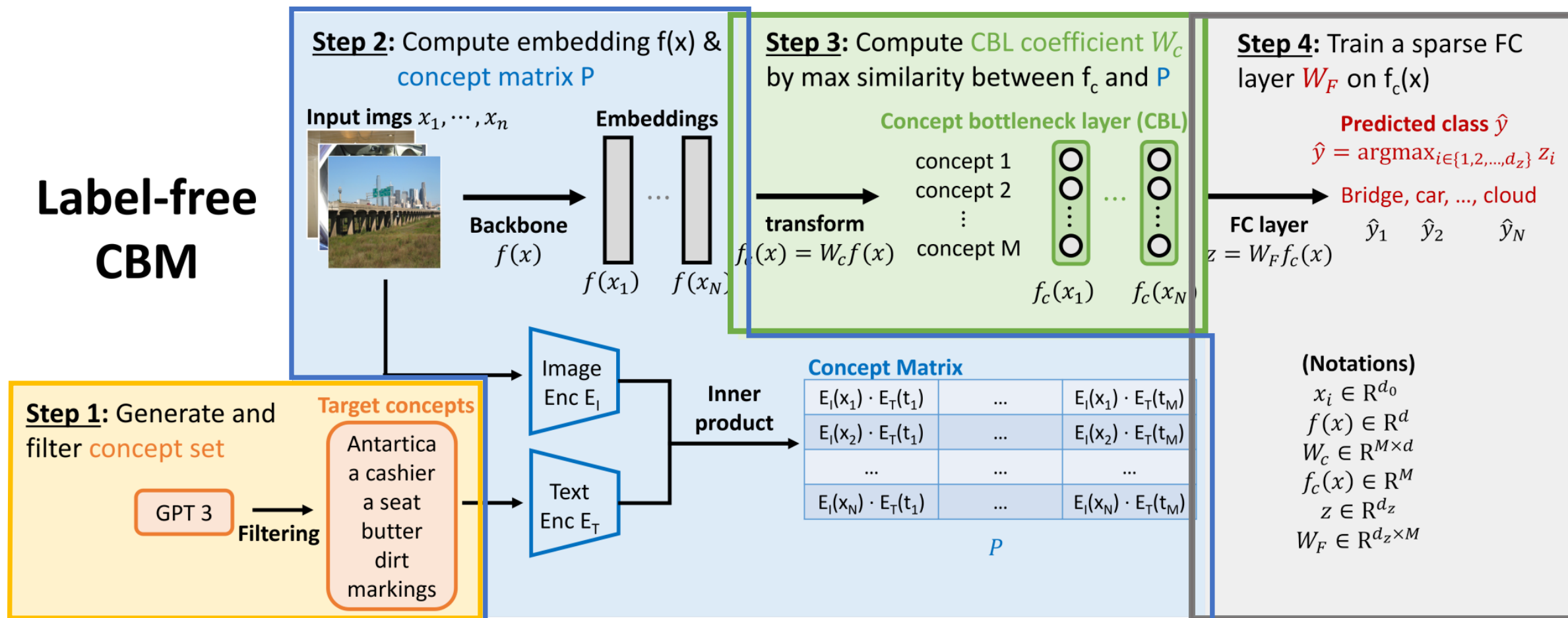
Label-free CBM

Transform any neural network into an **interpretable CBM without labeled concept data** while preserving accuracy comparable to the original neural network

- Use a LLM (GPT-3) to generate concept set
- Use CLIP-Dissect to align concepts with any backbone
 - No need of labeled concept data

Label-free CBM

Label-free CBM



Hand-ons time!



TCAV

Hand-ons time!



CBM

Hand-ons time!



Label Free CBM

Sources

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